

Gap-Bracket: Two-Layer Phylogenetic SCFG Parsing of RNA Consensus Structures with Pseudoknots

Ali Ahmadi Esfidi

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Abstract

We present Gap-Bracket, a method that predicts the consensus secondary structure of an RNA alignment, pseudoknots included. It combines the Knudsen–Hein (KH-99) stochastic context-free grammar with a phylogenetic model of unpaired and paired alignment columns, and parses the alignment into two nested layers: a first CYK pass predicts the nested structure, and a second pass predicts pseudoknotted pairs among the remaining columns at a tuned cost per pair. The substitution model and grammar are estimated from 337 Rfam families, and the decoding settings are tuned on 100 further families. The tuning method was selected by cross-validation grouped by Rfam clan. On a clan-disjoint test set of 246 Rfam alignments with literature-derived structures, 47 of them pseudoknotted, Gap-Bracket reaches a base-pair F1 of 0.759 (95% CI 0.723–0.792). That is statistically indistinguishable from RNAalifold (0.767) and IPknot (0.757). On pseudoknotted base pairs its F1 is 0.541, against 0.113 for IPknot. It detects a pseudoknot in 60% of pseudoknotted alignments at a specificity of 94%, and predicts an alignment in a median of 15 ms.

1 Introduction

RNA function depends on base pairing, which forms helical stems and loops. When several homologous sequences are aligned, compensatory mutations reveal which columns pair, so the consensus structure of an alignment can be inferred more reliably than the structure of one sequence. Pseudoknots, in which pairs cross, occur in many functional RNAs, but most consensus predictors either exclude them or predict them poorly.

KH-99 [7] and Pfold [8] combine a stochastic context-free grammar (SCFG) with a phylogenetic model of the alignment. SCFGs describe nested structures only. Gap-Bracket extends this approach to pseudoknots with a second parsing layer and asks whether such an extension is competitive. Our contributions are:

1. **A two-layer parser** with helix initiation and extension ratios, a pseudoknot pair ratio and an evidence

weight (Section 3);

2. **A benchmark** of 683 Rfam alignments with literature-derived structures, split by clan into training, tuning and test sets, scored with pseudoknot-specific metrics against RNAalifold and IPknot (Section 4);
3. **A tuning protocol** that chooses between tuning methods by clan-grouped cross-validation (Section 5);
4. **An evaluation** that separates overall accuracy, pseudoknot accuracy, ablations and running time, with bootstrap confidence intervals (Section 6).

2 Related Work

Consensus structure prediction. RNAalifold [2] from the ViennaRNA Package [9] folds an alignment by combining averaged free energies with a covariation term; it predicts nested structures. KH-99 and Pfold use an SCFG with phylogenetic column likelihoods, and PPfold [15] parallelises Pfold.

Pseudoknots. Exact dynamic programming for general pseudoknots, such as the algorithm of Rivas and Eddy [11], runs in $O(n^6)$ time; stochastic multiple context-free grammars [6] and pair stochastic tree adjoining grammars [10] model restricted classes at lower but still super-cubic cost. IPknot [13] decomposes a structure into nested levels and selects pairs from base-pairing probabilities by integer programming; it accepts alignments, which makes it the closest pseudoknot-capable baseline for our setting. Knotify [1] detects pseudoknots with syntactic pattern recognition, and deep-learning models such as E2Efold [3] and SPOT-RNA [14] predict pairing matrices from single sequences.

Gap-Bracket stays within cubic-time CYK. Like IPknot it builds pseudoknotted structures from nested layers, but it scores each layer with a phylogenetic SCFG rather than with thermodynamic pair probabilities.

3 Model

The input is an alignment of N sequences with n columns; the output is a set of base pairs between columns, drawn in dot-bracket notation. Table 1 lists the settings of the model.

Symbol	Setting	Value
ρ_{init}	helix initiation ratio	0.16
ρ_{ext}	helix extension ratio	1.59
ρ_{cross}	crossing ratio	1
ρ_{pk}	pseudoknot pair ratio	0.56
β	evidence weight	0.8
h_{min}	minimum hairpin loop	3

Table 1: Settings of Gap-Bracket. Both layers use the same ρ_{init} and ρ_{ext} ; the values are chosen in Section 5.

3.1 Structure grammar

The KH-99 grammar in Chomsky normal form is

$$\begin{aligned}
 S &\rightarrow L S \mid \$M E \mid s & L &\rightarrow \$M E \mid s \\
 F &\rightarrow \$M E \mid L S & \$M &\rightarrow B F \\
 B &\rightarrow d & E &\rightarrow d
 \end{aligned}$$

S generates a loop region of one or more elements, and L one element: an unpaired base s or a helix. $\$M E$ generates a base pair $d F d$, whose interior F is either another pair (stacking) or a loop of at least two elements. Every nested structure has exactly one derivation, so the maximum-likelihood rule probabilities are the relative rule frequencies in the training structures.

3.2 Phylogenetic column model

Unpaired columns evolve under a 4×4 rate matrix R , and paired columns under a 16×16 rate matrix over dinucleotides [7]. For every tree, Felsenstein’s pruning algorithm [4] gives $\log P_{\text{single}}(i)$ for each column and $\log P_{\text{pair}}(i, j)$ for each pair of columns, with gaps as unknown nucleotides [8]. Identical columns have identical likelihoods, so both are computed once per distinct column pattern.

Estimation. Each training family has the same total weight, split over its sequences. For ordered pairs of sequences with at least 85% identity over columns where both have a base, substitutions are counted in unpaired and paired columns (the latter symmetrised) and divided by the expected exposure,

$$R_{XY} = \frac{c_{XY}}{P_s P_X \sum_p t_p N_p}, \quad R_{XX} = - \sum_{Y \neq X} R_{XY},$$

where t_p is the tree distance and N_p the number of comparable columns of sequence pair p , P_X the frequency

of X and P_s the fraction of unpaired positions; the paired matrix uses $2/P_p$ in place of $1/P_s$.

Trees. Trees have a neighbour-joining topology [12] on Jukes–Cantor distances, with maximum-likelihood branch lengths under R (golden-section search per branch with inside and outside partial likelihoods), rooted at the midpoint. For estimation this is done in two stages: neighbour-joining branch lengths give a first model, whose rates then set the branch lengths for the final estimate.

3.3 Two-layer parsing

Extending the grammar with the column evidence gives the Viterbi recursions below; all terms are log-probabilities and q denotes rule probabilities.

$$\begin{aligned}
 \text{Pair}(i, j) &= M(i, j-1) + \log P_{\text{pair}}(i, j) + \chi(i, j) \log \rho_{\text{cross}} \\
 M(i, j) &= \log q_{\$M \rightarrow BF} + \max (F_{\text{st}}(i+1, j) + \log \rho_{\text{ext}}, \\
 &\quad F_{\text{lp}}(i+1, j) + \log \rho_{\text{init}}) \\
 F_{\text{st}}(i, j) &= \log q_{F \rightarrow \$ME} + \text{Pair}(i, j) \\
 F_{\text{lp}}(i, j) &= \log q_{F \rightarrow LS} + \max_k L(i, k) + S(k+1, j) \\
 L(i, j) &= \log q_{L \rightarrow \$ME} + \text{Pair}(i, j) \quad (i < j) \\
 S(i, j) &= \max (\log q_{\$S \rightarrow \$ME} + \text{Pair}(i, j), \\
 &\quad \log q_{\$S \rightarrow LS} + \max_k L(i, k) + S(k+1, j))
 \end{aligned}$$

For a single column, $S(i, i) = \log q_{\$S \rightarrow s} + \log P_{\text{single}}(i)$, and likewise for L . The helix initiation ratio ρ_{init} applies to a pair that closes a loop, i.e. the innermost pair of a helix, and the helix extension ratio ρ_{ext} to a pair stacked on another pair; with $\rho_{\text{init}} < \rho_{\text{ext}}$, helices are cheaper to extend than to start. Splitting F into its stacked and loop derivations applies the ratios exactly.

First layer. The parser runs on all columns and returns a nested structure.

Second layer. The parser runs again on the columns that the first layer leaves unpaired and returns a second nested set of pairs, which may cross the first. Each of its pairs is weighted by the pseudoknot pair ratio ρ_{pk} , and by ρ_{cross} once for every first-layer pair it crosses (χ counts these). With $\rho_{\text{pk}} < 1$ a pseudoknot helix must be better supported by the alignment than a nested one.

The two layers describe bi-secondary structures, the class IPknot predicts with two levels. The search is greedy: the second layer can only use columns the first leaves free. We also evaluated a *joint refinement* that re-parses each layer given the other and keeps a round only if the joint score of both layers increases (Section 5).

3.4 Evidence weight and hairpin loops

Every column log-likelihood is multiplied by the evidence weight β before parsing. The phylogenetic model treats columns as independent and sequences as related

only through the tree, which overstates the evidence of a small alignment; with $\beta < 1$ the grammar counts for more. A pair (i, j) is admitted only if $j - i > h_{\min}$ in alignment coordinates, with $h_{\min} = 3$; the bare grammar allows loops of two columns.

Output. The predicted pairs are drawn with the largest nested subset as $()$, the largest nested subset of the rest as $[]$, and so on (exact interval dynamic programming). This affects only the drawing, not the pairs or any score.

Implementation and verification. Each span length of the chart is filled in one vectorised step, and the bifurcation reads contiguous array slices, so a pass costs $O(n^3)$ arithmetic with $O(n)$ interpreter iterations. The CYK recursions (for $h_{\min} = 2$ and 3), the structure scoring used by refinement, the largest-nested-subset program and pattern-compressed likelihoods are all tested against exhaustive enumeration or direct computation.

4 Benchmark and Protocol

4.1 Data

The benchmark is built from Rfam 15.1 [5] seed alignments. Only the 769 families whose consensus structure is annotated as published or taken from PseudoBase [16] are used. Structures annotated as predicted (by RNAalifold, CMfinder and similar tools) are excluded, because they would reward agreement with the tool that made them. Families of one Rfam clan are kept together, and clans are shuffled with a fixed seed within two strata (with or without a pseudoknotted family) and dealt 50/15/35% to training, tuning and test.

Training families keep up to 100 seed sequences. Each tuning and test family contributes one alignment of 4–12 random seed sequences, the setting in which a user supplies a small alignment. In every alignment, columns where at most half of the sequences have a base are removed together with the reference pairs that use them; alignments with fewer than three reference pairs, and tuning or test alignments longer than 400 columns, are skipped (Table 2).

Set	Aln.	PK	Clans	Seqs	Columns
Training	337	66	316	4–100	23–1208
Tuning	100	22	94	4–12	19–307
Test	246	47	216	4–12	20–383

Table 2: Benchmark sets: alignments, alignments with a pseudoknot, clan groups, sequences and columns per alignment. Median test alignment: 7 sequences, 94 columns.

4.2 Metrics

- **All base pairs:** positive predictive value (PPV), sensitivity and F1, pooled over the alignments of a set.
- **Pseudoknotted pairs:** the same measures restricted to pairs that cross at least one other pair of their own structure. This definition covers both helices of a pseudoknot and does not depend on bracket layers.
- **Pseudoknot detection:** per alignment, sensitivity (a pseudoknot is predicted when the reference has one) and specificity (none is predicted when the reference has none).

Confidence intervals come from 10,000 bootstrap re-samples of alignments; differences between methods use paired resamples.

4.3 Baselines

- **RNAalifold 2.7.2**, consensus minimum free energy, default parameters (nested structures only).
- **IPknot 1.1.0** in aligned mode, mixing averaged McCaskill and RNAalifold base-pairing probabilities as recommended for alignments, with default thresholds (pseudoknots allowed).
- **KH-99 CYK:** our parser with one layer, all ratios 1, $\beta = 1$ and $h_{\min} = 2$.

Knotify, SPOT-RNA and E2Efold predict from single sequences and would need a rule for combining per-sequence predictions into a consensus. They were not evaluated.

4.4 Estimated parameters

The 337 training families give the parameters in Table 3. GC/CG pairs make up 49.7% of paired columns (AU/UA 35.0%, GU/UG 10.0%). Transitions are faster than transversions, and in paired columns, changes that keep a pair canonical, such as UG→CG, dominate. The grammar rule probabilities are $S \rightarrow LS \mid \$ME \mid s$: 0.896, 0.013, 0.092; $L \rightarrow \$ME \mid s$: 0.077, 0.923; and $F \rightarrow \$ME \mid LS$: 0.808, 0.192, so most pairs are stacked on another pair.

5 Tuning

5.1 Tuning methods

Seven settings are tunable: ρ_{init} and ρ_{ext} of each layer, ρ_{cross} , ρ_{pk} and β . Ratios take values $2^{k/3}$ for $k = -8, \dots, 8$ (0.16–6.35), $\rho_{\text{pk}} \in \{10^{-k/4} : k = 0, \dots, 8\}$ (1–0.01) and $\beta \in \{0.4, 0.5, \dots, 1.1, 1.25\}$; $h_{\min} = 3$ is

X / Y	AU	UA	GC	CG	UG	GU
AU	−1.70	0.13	0.43	0.10	0.02	0.60
UA	0.13	−1.70	0.10	0.43	0.60	0.02
GC	0.30	0.06	−1.19	0.08	0.01	0.45
CG	0.06	0.30	0.08	−1.19	0.45	0.01
UG	0.08	2.12	0.04	2.31	−4.98	0.03
GU	2.12	0.08	2.31	0.04	0.03	−4.98

Table 3: Paired-column substitution rates between canonical pairs, estimated from the training set (the diagonal includes rates to non-canonical pairs).

fixed. The objective is pooled base-pair F1 on the tuning set, with ties broken towards neutral values. We compare three tuning methods.

- **Shared ratios, exhaustive search:** both layers share ρ_{init} and ρ_{ext} and $\rho_{\text{cross}} = 1$; the full grid of the remaining four settings ($17 \times 17 \times 9 \times 9$ candidates) is scored, so the optimum is global.
- **Separate ratios, coordinate ascent:** all seven settings, one at a time from neutral.
- **Separate ratios, genetic algorithm:** all seven settings; population 32, tournament selection, uniform crossover, mutations of one or two grid steps and elitism, followed by coordinate ascent.

Base-pair counts are cached for every candidate and alignment and computed in parallel; the protocol below scores 31,830 candidates in 17 minutes on a laptop.

5.2 Selecting the method

The tuning set is split into 10 folds by clan. Each method is tuned on nine folds and predicts the tenth, and the pooled held-out predictions give its cross-validated F1 (Table 4). The method with the highest estimate is chosen, with ties going to fewer settings, and is then fitted on the whole tuning set.

Tuning method	Held-out F1		Picks
	2 layers	+ refinement	
None (all settings 1)	.728	.731	—
Separate, coordinate	.745	.759	6
Separate, genetic	.755	.754	9
Shared, exhaustive	.755	.755	5
with $\beta = 1$	.753	—	4
<i>Untuned baselines on the same alignments</i>			
RNAalifold	.758		
IPknot	.757		

Table 4: Clan-grouped 10-fold cross-validation on the tuning set. *Picks*: distinct settings chosen over the folds (two layers).

Tuning raises held-out F1 from 0.728 to 0.755. On this benchmark the three methods differ by at most 0.010. Joint refinement does not change the chosen method’s estimate and quadruples tuning time (70 minutes), so it is not used. Exhaustive search over shared ratios ties with the genetic algorithm using three fewer settings and is selected. The final settings (Table 1) keep $\rho_{\text{init}} < \rho_{\text{ext}}$ and charge each pseudoknotted pair a ratio of 0.56.

6 Results

All results in this section are on the test set, which was scored once after the model, tuning method and settings had been fixed.

6.1 Accuracy

Table 5 summarises the test results.

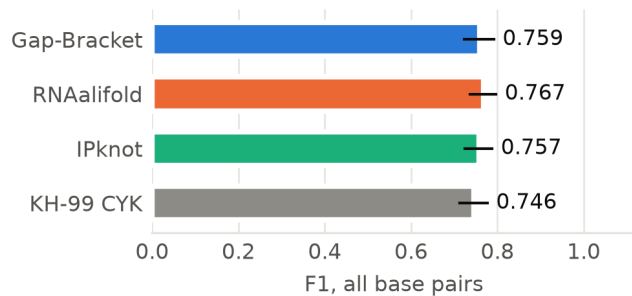


Figure 1: Test-set F1 on all base pairs, with 95% bootstrap intervals.

All base pairs. Gap-Bracket reaches F1 0.759 [0.723, 0.792] (Figure 1). The differences to RNAalifold (−0.008, 95% CI −0.034 to +0.015) and IPknot (+0.002, −0.019 to +0.022) are not significant. Gap-Bracket is more precise than RNAalifold (PPV 0.802 vs 0.789) and less sensitive (0.721 vs 0.747).

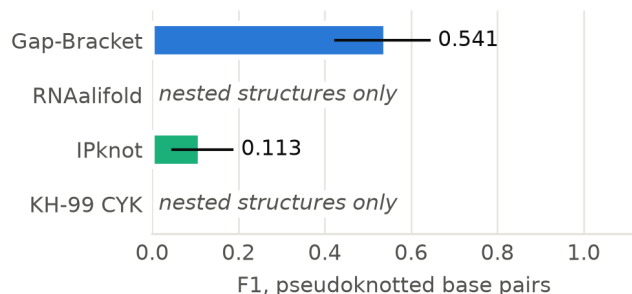


Figure 2: Test-set F1 on pseudoknotted base pairs, with 95% bootstrap intervals. RNAalifold and KH-99 CYK cannot predict pseudoknots.

Pseudoknots. On pseudoknotted pairs Gap-Bracket reaches F1 0.541 [0.425, 0.642] (Figure 2), against

Method	All base pairs			Pseudoknotted pairs			PK detection		F1 by alignment	
	PPV	Sens.	F1	PPV	Sens.	F1	Sens.	Spec.	PK (47)	Nested (199)
KH-99 CYK	.791	.706	.746	–	–	–	–	–	.700	.761
RNAalifold	.789	.747	.767	–	–	–	–	–	.709	.785
IPknot	.803	.716	.757	.140	.094	.113	.319	.809	.707	.773
Gap-Bracket	.802	.721	.759	.596	.496	.541	.596	.940	.771	.755

Table 5: Test-set accuracy (246 alignments). *F1 by alignment*: overall pair F1 on the 47 alignments whose reference has a pseudoknot and on the 199 without one. Dashes: the method cannot predict pseudoknots.

0.113 [0.048, 0.186] for IPknot, a difference of +0.429 (+0.297 to +0.541). It predicts a pseudoknot in 60% of the pseudoknotted alignments and in 6% of the others; IPknot does so in 32% and 19%. On pseudoknotted alignments, Gap-Bracket has the highest overall F1 (0.771 vs 0.709 for RNAalifold). On alignments without pseudoknots, RNAalifold is best (0.785 vs 0.755).

6.2 Ablations

Variant	All pairs		PK pairs	
	F1	Δ	F1	Δ
Gap-Bracket	.759		.541	
First layer only	.747	-.012*	–	
$h_{\min} = 2$	.756	-.003*	.528	-.013
$\beta = 1$ (retuned)	.764	+.005	.506	-.036
Separate, coordinate	.758	-.001	.538	-.003
Separate, genetic	.760	+.001	.549	+.008

Table 6: Test-set ablations and settings from the rejected tuning methods. Δ : difference to Gap-Bracket; *: 95% paired bootstrap interval excludes zero.

The second layer adds 0.012 overall F1, with a confidence interval excluding zero, and provides all of the pseudoknot accuracy (Table 6). The minimum hairpin loop adds 0.003, also significant. The evidence weight trades 0.005 overall F1 for 0.036 pseudoknot F1; neither difference is significant. The settings of the rejected tuning methods agree with Gap-Bracket to within 0.008 on both measures, consistent with the cross-validation.

6.3 Running time

Gap-Bracket needs a median of 14.5 ms per test alignment (maximum 231 ms), RNAalifold 11.3 ms (374 ms) and IPknot 116 ms (4.6 s); the 246 alignments take 6.5 s, 5.7 s and 92 s (Figure 3). The RNAalifold and IPknot times include starting a process per alignment.

7 Discussion

What the second layer buys. A nested SCFG cannot represent pseudoknots, and adding a second layer

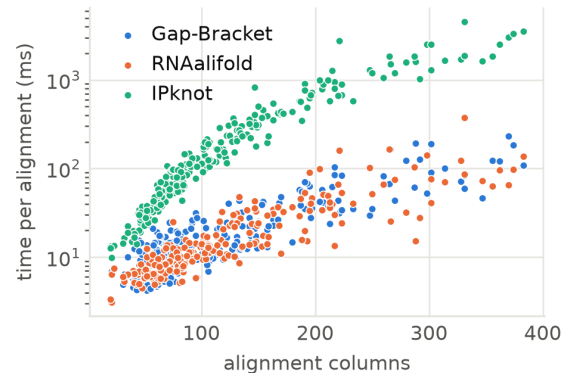


Figure 3: Time per test alignment against alignment length (tree, likelihoods and both layers for Gap-Bracket). All methods ran with 8 parallel workers on an Apple-silicon laptop.

with a cost per pseudoknotted pair recovers half of the pseudoknotted pairs at a PPV of 0.60, without lowering overall accuracy. IPknot, which uses thermodynamic pair probabilities and per-level thresholds, rarely predicts pseudoknots on these alignments of 4–12 sequences.

Where it falls short. On alignments without pseudoknots, RNAalifold’s energy model is more accurate than the phylogenetic SCFG (0.785 vs 0.755), and Gap-Bracket still predicts a pseudoknot for 6% of them. The KH-99 grammar has no loop-length or stacking-energy terms, and its substitution model treats sequences as related only through the tree.

Limitations.

- *References.* Rfam consensus structures, even literature-derived ones, are curated alignments rather than solved three-dimensional structures, and they may miss or misplace pseudoknots.
- *Pseudoknot model.* The second layer is a greedy decomposition into two nested layers, not a pseudoknot grammar. It cannot represent structures that need three layers, and it can miss a pseudoknot whose helices are recognisable only together.

Joint refinement of both layers did not help on this benchmark.

- *Baselines.* Single-sequence pseudoknot predictors, such as Knotify, SPOT-RNA and E2Efold, were not evaluated, and IPknot was run with its default aligned configuration only.
- *Statistics.* With 47 pseudoknotted test alignments, pseudoknot measures have intervals about ± 0.1 wide.

Alternatives not adopted. Earlier development on a smaller data set also examined posterior (maximum-expected-accuracy) decoding, minimum stem lengths, a separate weight for unpaired columns and a smoothed tuning objective; none gave a reliable cross-validated gain.

8 Conclusion

Gap-Bracket extends a phylogenetic SCFG to pseudoknots with a second parsing layer, a pseudoknot pair ratio and an evidence weight, all tuned by a cross-validated protocol on clan-disjoint Rfam data. On 246 held-out alignments it matches RNAalifold and IPknot on overall base-pair accuracy, predicts pseudoknotted pairs with F1 0.541 against 0.113 for IPknot, and runs about eight times faster than IPknot. The remaining gap to energy-based folding on nested structures points to richer loop models as the next step.

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